Thomas Raymond Deatherage

thomas.deatherage@pm.me | 540-845-8682 | Georgia, USA | thomasdeatherage.com

Education

Georgia Institute of Technology | *Expected 2025*

- MS Computer Science, Computing Systems Specialization (Project Track)
- GPA: 4.0

Virginia Commonwealth University | *2008 - 2012*

- BS Biochemistry, Mathematics Minor
- Honors College
- GPA: 3.92 (summa cum laude)

Professional Development

University of Chicago Basic Program | *2023 - Present*

- Four year liberal arts curriculum that studies the foundational texts in Western political and social thought.

Current Research

Research Assistant, BioVision Lab / *Georgia Institute of Technology & Florida Natural History Museum Collaboration / June 2024 - Present*

- Developing a novel research platform for ecologists and other scientists using Machine Learning, Computer Vision, and Large Language Models
- Investigating novel approaches to species classification focusing on the challenges of the biodiverse 'longtail' of species data

Professional Experience

Senior Software Engineer / *Meta / March 2022 - July 2023*

- Worked on systems architecture, focusing on issues of scale
- Developed product ideas from an engineering perspective
- Served as technical lead and architect of five separate project groups, primarily focusing on chat and telephony applications

Senior Software Engineer / *Kustomer / July 2018 - March 2022*

- Engineered a distributed microservices backend architecture
- Focused on issues of robustly handling traffic and storage scale
- Drove product direction from a technical perspective

Co-Founder / *Funderdog / April 2017 - July 2018*

- Technical co-founder of an online donations social network
- Designed and implemented the entire codebase

Software Engineer / *Shutterstock* / *September 2013 - April 2017*

- Developed the internal payments portal
- Developed the internal business-insights tools
- Developed one of *Shutterstock's* niche brands, *Offset*, a high-end photography provider

Associate Scientist / *Arista Laboratories* / *November 2012 - May 2013*

- Performed chemical constituent analysis in both a "wet" and "dry" lab setting

Teaching Assistant / *Virginia Commonwealth University* / *January 2011 - May 2011*

- Instructed General Chemistry Introduction course
- Led small-group recitation sessions
- Proctored exams

Awards and Achievements

- Winner, AngelHack Manhattan Hackathon (2018)

Research Interests

- Computer Vision and Machine Learning
- Large Language Models and Vision Language Models
- Computing Applications in Ecology, Biology, Biochemistry and Psychology
- Human-Computer Interaction (HCI) and Human-Centered Computing (HCC)

Technical Skills

- **Programming Languages**: JavaScript/TypeScript, Python, Ruby, Perl, C, Java
- **Machine Learning & Data Analysis**: PyTorch, Pandas, SciPy, NumPy
- **Research Areas**: Computer Vision, HCI, Machine Learning

Publications & Works in Progress

- "[Title TBD]: Classifying the Long-tail of Diverse Species" (Expected 2025)
  - Research focusing on novel approaches to species classification in ecological datasets
  - Addressing challenges in long-tail distribution of biological data